

Machine Learning Basics

Machine Learning is a branch of Artificial Intelligence that enables systems to learn from data and improve performance without being explicitly programmed. Types of Machine Learning **Supervised Learning:** Uses labeled data to train models. **Unsupervised Learning:** Finds hidden patterns in unlabeled data. **Reinforcement Learning:** Learns through rewards and penalties. Common Algorithms Linear Regression Logistic Regression Random Forest Decision Trees Support Vector Machines Neural Networks Applications of Machine Learning Healthcare Prediction Recommendation Systems Fraud Detection Chatbots and Virtual Assistants Computer Vision Deep Learning Deep Learning is a subset of Machine Learning that uses neural networks with multiple layers to process complex patterns in large datasets. Natural Language Processing NLP allows machines to understand, process, and generate human language. Generative AI Generative AI uses large language models to generate text, images, code, and other content based on prompts.